How long an outburst lasts helps forecast when the next one begins

SS Cygni: a century of amateur observations, and forecasts fixed before the test data were opened

① THE STAR

companion star — disc around a white dwarf

About every seven weeks the disc flares, then fades: an outburst

Real nightly brightness, 100 days of 1960

Does an outburst's length help forecast when the next one starts?

② THE TEST

Two forecasters
each gives odds on when the next outburst starts

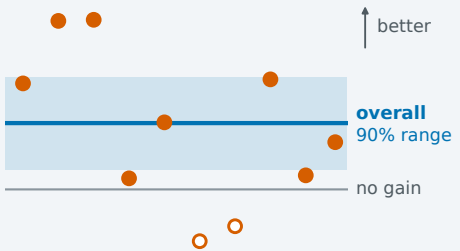
recent quiet spells

recent quiet spells + outburst length

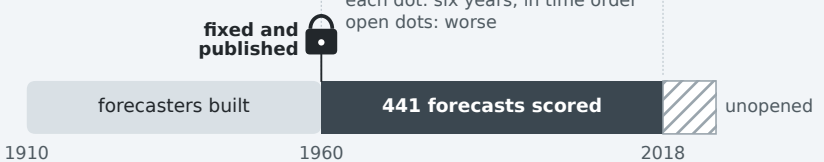
expects a 9% longer wait for each doubling of length

③ THE ANSWER

Better overall, and in 8 of 10 six-year blocks
a small gain, but its 90% range stayed above zero, as required in advance



each dot: six years, in time order
open dots: worse



Other stars Ten other dwarf novae: their early records were too patchy to repeat the test.

Next Score the fixed forecasts on SS Cygni's observations from 2018 onward, still unopened.